Project Title	**Rising Waters: Machine Learning Based Flood Prediction System**		
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8. Problem Solution Fit

The Problem-Solution Fit canvas checks whether the proposed solution genuinely matches the problem and the emotional/practical context the user is in, rather than simply being a technically interesting thing to build. Framing it this way surfaced the constraint that most directly shaped the frontend: the target user cannot be assumed to be technically fluent, so the interface has to do the interpretive work the raw model output cannot.

Aspect	Description
Problem	No accessible, data-driven tool exists to convert raw meteorological data into an immediate flood-risk classification for residents/authorities.
Solution	A Flask-served ML classifier trained on historical rainfall and weather data, returning an instant probability, severity tier, and actionable guidance.
Trigger to Act	Onset of monsoon season, forecast of heavy rainfall, or general seasonal preparedness checks.
Emotional Response	Reduced uncertainty; a sense of preparedness rather than reactive concern.
Customer Constraints	Users may not have technical expertise; the interface must interpret results in plain language rather than raw probabilities alone.

This constraint is the direct reason the /predict route in app.py does more than return a number: it always accompanies the probability with a severity label, a list of plain-language insights, and a matching list of recommended protocols, so that no output ever reaches the user as an unexplained statistic.